TASK 2: REST API Data Fetcher & Dashboard

Domain: API Integration | Level: Basic

What is this project?

This project fetches live data from a public REST API using Java's built-in HttpClient, parses the JSON response, displays it as a formatted table in the console, and handles errors like timeouts and invalid responses.

Key Concepts to Learn First

• HTTP GET request using java.net.http.HttpClient (Java 11+)

• JSON parsing using org.json library

• Error handling with try-catch (IOException, InterruptedException)

• Formatting console output as a table

• API rate limits awareness

Step-by-Step Explanation

Step 1 – Choose a Public API

We use JSONPlaceholder (free, no key needed): https://jsonplaceholder.typicode.com/posts

Step 2 – Send HTTP GET Request

Use HttpClient.newHttpClient() and HttpRequest.newBuilder().uri(...).GET().build()

Step 3 – Read the Response

Use HttpResponse and get the body as a String

Step 4 – Parse JSON

Use org.json.JSONArray and JSONObject to extract fields

Step 5 – Display as Table

Print headers and rows with printf for alignment

Step 6 – Handle Errors

Wrap everything in try-catch; check response status code != 200

Complete Java Code

```
import java.net.URI;
import java.net.http.*;
import java.time.Duration;
import org.json.*;

/**
 * REST API Data Fetcher
```

```
 * Fetches posts from JSONPlaceholder and displays them as a table.
 * Dependency: org.json (add json-20231013.jar to classpath)
 */
public class APIFetcher {

    // Public API URL – no API key needed
    static final String API_URL = "https://jsonplaceholder.typicode.com/posts";

    public static void main(String[] args) {
        // ■■ Step 1: Create HTTP Client with timeout ■■
        HttpClient client = HttpClient.newBuilder()
            .connectTimeout(Duration.ofSeconds(10))
            .build();

        // ■■ Step 2: Build GET Request ■■
        HttpRequest request = HttpRequest.newBuilder()
            .uri(URI.create(API_URL))
            .GET()
            .header("Accept", "application/json")
            .timeout(Duration.ofSeconds(15))
            .build();

        try {
            // ■■ Step 3: Send Request and Get Response ■■
            HttpResponse<String> response =
                client.send(request, HttpResponse.BodyHandlers.ofString());

            // ■■ Step 4: Handle HTTP errors ■■
            int statusCode = response.statusCode();
            if (statusCode != 200) {
                System.err.println("Error: Server returned status " + statusCode);
                return;
            }

            // ■■ Step 5: Parse JSON Response ■■
            JSONArray posts = new JSONArray(response.body());

            // ■■ Step 6: Display as formatted table ■■
            System.out.println("=== POSTS FROM JSONPlaceholder API ===");
            System.out.printf("%-5s %-10s %-40s%n", "ID", "UserID", "Title");
            System.out.println("-".repeat(60));

            // Show first 10 posts only
            for (int i = 0; i < Math.min(10, posts.length()); i++) {
                JSONObject post = posts.getJSONObject(i);
                int id     = post.getInt("id");
                int userId = post.getInt("userId");
                String title = post.getString("title");

                // Truncate long titles for neat display
                if (title.length() > 37) title = title.substring(0, 37) + "...";

                System.out.printf("%-5d %-10d %-40s%n", id, userId, title);
            }

            System.out.println("\nTotal posts fetched: " + posts.length());

        } catch (java.net.http.HttpTimeoutException e) {
            System.err.println("Error: Request timed out. Check internet connection.");
```

```
        } catch (java.io.IOException e) {
            System.err.println("Error: Network problem – " + e.getMessage());
        } catch (InterruptedException e) {
            System.err.println("Error: Request was interrupted.");
            Thread.currentThread().interrupt();
        } catch (JSONException e) {
            System.err.println("Error: Could not parse JSON – " + e.getMessage());
        }
    }
}
```

Expected Output

```
=== POSTS FROM JSONPlaceholder API ===
ID    UserID     Title
------------------------------------------------------------
1     1          sunt aut facere repellat provident occaeca...
2     1          qui est esse
3     1          ea molestias quasi exercitationem repellat ...
...
Total posts fetched: 100
```

How to Compile & Run

```
# Download org.json jar:
# https://mvnrepository.com/artifact/org.json/json

# Compile:
  javac -cp ".;json-20231013.jar" APIFetcher.java   (Windows)
  javac -cp ".:json-20231013.jar" APIFetcher.java   (Mac/Linux)

# Run:
  java -cp ".;json-20231013.jar" APIFetcher          (Windows)
  java -cp ".:json-20231013.jar" APIFetcher          (Mac/Linux)
```

TIP: You'll learn HTTP requests • JSON parsing • Error handling • Table formatting